Daniel Hardesty Lewis

Education

expected 2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY

2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

2017 **Certificate, Scientific Computation**, *University of Texas*, Austin, TX

Experience

Professional

Apr. 2025 – Present **Co-founder**, *Properlytic*, New York, NY

Targetting the last blue ocean in consumer real estate technology: individual home price forecasting

- Here's our pitch
- Built out a current home price prediction ML model, achieving 12% MAPE against Zillow's 8.4% in the hardest US market to do it: Manhattan
- That model stack is identical to some of the recent cutting-edge physics foundation models, using a Variational AutoEncoder (VAE) approach
- Uses a data-driven approach to auto-configure the vast majority of hyperparameters and even some of the model architecture, trains in under 7 min
- Interest from a16z & MetaProp

Nov. 2024 – Present **Co-founder**, *PoliBOM*, Seattle, WA

Top-5% YC entrant, KPMG rolled out their Tariff Modeller after we put polibom.com in front of them

- Now targetting early stage investment

Nov. 2023 – Present **Founder**, *Summit Geospatial*, New York, NY

Developed the highest quality seamless elevation data for Texas, summitgeospatial.com

- Now targetting recent interest received to license to AI labs such as OpenAI & Cohere

Aug. 2021 – Aug. 2023 **Senior Data Scientist & Technical Lead**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Spun out Summit Geospatial alongside UT's VPs of IP
- Led a principal project of the $40M Texas Disaster Information System (TDIS) disaster resiliency initiative
- Contributed research towards Paola Passalacqua's 2022 EGU Bagnold Medal, one of geomorphology's highest honors
- Scaled climate and flood models on world-leading supercomputers, utilizing millions of dollars in computational resources and million-node jobs
- Evaluated and accepted tens of millions of dollars in compute requests
- Managed relationships and developed projects with federal, state agencies, and universities including NOAA, NSF, US Army Corps of Engineers, Texas General Land Office, USC, UT, A&M

Feb. 2018 – Jul. 2021 **Data Scientist & Research Engineer**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Competed in DARPA World Modelers program, focusing on disaster resiliency in East Africa
- Research received a mention from President of AAAI Yolanda Gil during her farewell speech
- Helped teach Petrobras deep learning on HPC

Research

May 2025 – Present **Research assistant to Dir. Electrical Engineering Zoran Kostic**, *Columbia University*, New York, NY

Targetting theory publications to DeepSeek's GRPO algorithm

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Jan. 2025 – Present
Research assistant to Dir. Financial Engineering Ali Hirsa, *Columbia University*, New York, NY
Achieved better than commercial R^2 on backtested holdout prediction of daily mutual fund returns on a variety of asset classes
- Using a DL latent factor model to produce better factors than our benchmark: Fama-French's 5-factor model
- Now targetting SHAP explainability publication

July 2019 – Aug. 2019
Summer research intern to Pres. AAAI Yolanda Gil, *University of Southern California*, Los Angeles, CA
Integrated hydrological models into team's DARPA-supported scientific model orchestration portal

Teaching

Jan. 2018 – Dec. 2020
Co-instructor, *The University of Texas at Austin*, Austin, TX
Helped teach graduate-level courses: *Machine Learning for the Geosciences*, *Intelligent Systems for the Geosciences*, and *Scientific Computation*, including C++, CUDA, HPC, & optimization

Key Projects

oppsAlert	Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15000+ points of individual homeowner protest data provided by the City of Austin
Stealth, RegTech	*Seeking PE investment:* Teaming up with a top-proptech operator for a very large opportunity to aid cities achieve permitting compliance via simple pipeline collating publicly available imagery against permitting data
Flood Hazard Assessment	Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.
Texas Disaster Information System (TDIS)	Led development of the core elevation data layer for a web-based spatial data system that supports resilient decision-making at the state and local levels.
Real-Time Inundation Mapping	Developed computationally efficient methods to produce high-resolution (1m) flood inundation maps from National Water Model outputs for emergency response personnel.

Technology Skills

Programming Languages	Python, Bash, PL/pgSQL, Fortran, Perl, C++	Operating Systems	GNU/Linux, macOS, Windows
Libraries & Frameworks	GDAL, CUDA, GeoPandas, RasterIO, NumPy, Plotly, Dash	Database Systems	PostgreSQL, Redis, Supabase, PostGIS
Scientific Software	MODFLOW, HAND, HEC-RAS, MESA	Cloud & Virtualization	AWS, Azure, Docker, Singularity
AI tooling	ChatGPT, Claude, Gemini, VSCode, v0.dev	ML	VAE, Tree-based models
Clustering techniques & metrics	DBSCAN, Hierarchical, Silhouette score, Calinski-Harabasz, Davies-Bouldin	Statistical & ML metrics	R^2, MAPE, RMSE, KL-div, Reconstruction loss

Languages

Spanish	Native / Bilingual Proficiency	French	Professional Working Proficiency

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